

A diagnostic-driven multi-agent Socratic teaching system

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Abstract

To elevate AI tutors from simple chatbots to genuine pedagogical partners, this study introduces a novel diagnosis-driven methodology for adaptive Socratic teaching. Current LLM-based tutors often lack the deep diagnostic capabilities required for authentic Socratic dialogue, resulting in superficial interactions. Our approach overcomes this by tightly coupling a real-time, multi-dimensional student cognitive model with the dialogue generation process. This methodology is implemented within a multi-agent system where: a Planner Agent first dynamically structures learning paths from raw documents; a Diagnoser Agent continuously assesses student mastery to create precise cognitive reports; and a Tutor Agent leverages these reports to conduct targeted, high-quality Socratic questioning. Experimental results demonstrate the superiority of our diagnosis-driven method. In blind comparisons, Our system achieved an average win rate of 82% against baseline systems, showing decisive advantages in "Socratic style" and teaching effectiveness. This study confirms that a deep diagnostic mechanism is the key to unlocking personalized and effective Socratic teaching in AI tutors.

CCS Concepts

• **Applied computing** → **Computer-assisted instruction; Interactive learning environments.**

Keywords

Intelligent Tutoring Systems, Socratic Teaching, Multi-Agent Systems, Diagnosis-Driven, Retrieval-Augmented Generation

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1 Introduction

With the rise of large language models, replicating the effectiveness of one-to-one human tutoring through Socratic teaching has become a central goal in personalized education[3]. However, authentic Socratic teaching is more than a series of "why" questions; it is a pedagogical art driven by deep, real-time diagnosis of a student's cognitive state. Most existing AI tutors, even those built on powerful language models, struggle to achieve this. Lacking an independent, structured diagnostic mechanism, their interactions often devolve into superficial question-answering, failing to provide the targeted, actionable guidance central to formative assessment and genuine Socratic inquiry[13]. The core challenge, therefore, is not merely simulating a Socratic style, but enabling Socratic reasoning—a process that requires an AI tutor to dynamically plan instruction, precisely diagnose student understanding, and adapt its questioning strategy accordingly.

In recent years, emerging technologies such as large language models (LLMs) and retrieval-augmented generation (RAG) have shed new light on overcoming these obstacles. RAG enables systems to dynamically synthesize knowledge from unstructured documents, offering a potential solution to the long-standing "knowledge bottleneck" in AI development. Meanwhile, the powerful natural language capabilities of LLMs pave the way for constructing deep, flexible teaching dialogues that can adapt to diverse educational scenarios[15]. However, simply embedding these technologies into the traditional, tightly-coupled four-module architecture remains limited, as this design continues to restrict system scalability and maintainability.

To address these challenges, this study adopts and extends the modular design philosophy of Multi-Agent Systems (MAS)[12], proposing a diagnosis-driven, multi-agent adaptive tutoring framework. This framework represents a modernized, modular reconstruction of the traditional ITS architecture. By decoupling macro-level planning, micro-level diagnosis, and instructional execution, the framework significantly enhances both flexibility and scalability. It is composed of four core agents working collaboratively:

- The Planner Agent dynamically constructs knowledge bases and learning paths from unstructured documents, following a "divide-and-integrate" paradigm.

- The Diagnoser Agent performs real-time qualitative analysis during dialogue, maintaining a multidimensional cognitive model of the student to precisely assess their mastery.
- The Tutor Agent, guided by diagnostic reports, engages the student in high-quality Socratic dialogue.
- The Assessment Engine establishes “mastery validation checkpoints” at key stages to quantitatively verify learning outcomes, which supports a rigorous Socratic teaching process.

The central research question of this study is: How can we design and implement an intelligent tutoring system that ingests arbitrary unstructured materials, dynamically constructs an internal knowledge representation, and on this basis engages students in high-quality, adaptive Socratic dialogue through a closed-loop teaching process?

Experimental results demonstrate that the proposed framework significantly outperforms baseline methods in instructional planning, with higher recall in knowledge base construction and greater logical consistency in learning paths. In evaluating adaptive teaching capabilities, A/B blind testing against a baseline system lacking a diagnostic module showed that our system achieved an average win rate of 82% across multiple LLM-based AI evaluations, with decisive advantages in the “Socratic style” dimension. These findings confirm that a diagnosis-driven approach is fundamental to enhancing the personalization and adaptability of AI tutors, offering a robust architectural blueprint for the next generation of intelligent tutoring systems.

1.1 Problem statement and research objective

The core question of this study is: How to design and implement an intelligent teaching system that can receive any unstructured learning materials, dynamically construct its internal knowledge representation, and on this basis, conduct high-quality and adaptive Socratic dialogue with students? In order to solve this core problem, this study sets the following specific objectives:

- Design a modular and scalable multi-agent collaboration architecture in which the division of responsibilities of each agent can support the complex Socratic teaching process.
- A GraphRAG inspired dynamic knowledge base construction algorithm is implemented, which can automatically extract, integrate and structure core knowledge points from original documents.
- A hybrid intelligent path planning algorithm is developed, which combines the efficiency of vector retrieval with the reasoning ability of LLM to generate a logically coherent learning path for users’ learning goals.
- A ‘mastery validation checkpoint’ mechanism is introduced to verify the effectiveness of the Socratic dialogue and ensure students achieve reliable mastery.
- Implement a multidimensional dynamic student model that provides deep insight into Socratic dialogue and can track student ability over time.

1.2 Contributions of this paper

The main contributions of this paper can be summarized as the following three points:

- A dynamic teaching system architecture based on RAG and supporting Socratic teaching is proposed. The four core agents (planner, diagnostician, lecturer and Assessment Engine) are organically combined to realize the whole process automation from knowledge acquisition to personalized teaching.
- A novel knowledge base construction and path planning algorithm suitable for teaching scenarios is designed and implemented. Through the “divide and conquer-integrate” strategy and the “three-stage rocket” planning model, the problem of teaching planning on dynamic knowledge system is effectively solved.
- A “diagnosis-teaching” collaborative model is constructed and verified, in which the diagnostician’s in-depth analysis provides the necessary data support for the lecturer to conduct high-quality, non-scripted Socratic questioning, and the teaching effect is ensured by the “mastery verification checkpoint”.

1.3 Structure of the paper

The remainder of this article is organized as follows:

- Chapter 2 will review the work on intelligent teaching systems, student modeling and Socratic dialogue systems, and clarify the positioning of this research.
- Chapter 3 will elaborate the multi-agent system architecture, dynamic knowledge base construction algorithm, learning path planning model and the specific implementation of the “learn-practice-test-evaluate” integrated teaching cycle proposed in this paper.
- Chapter 5 will design and implement a series of experiments to evaluate the performance of the system in terms of knowledge base construction quality, path planning accuracy, and effectiveness of Socratic dialogue teaching.
- Chapter 6 summarizes the work of this paper and looks forward to future research directions.

2 Related Work

This chapter will review several core areas related to this research, including the architecture of traditional intelligent teaching system (ITS), student modeling and knowledge representation methods, and the latest applications of large language model (LLM) in education, so as to clarify the technical positioning and innovation of this research.

2.1 Intelligent Instructional System (ITS)

Intelligent Instructional Systems (ITS) are designed to simulate personalized one-on-one tutoring through artificial intelligence technology. The classic ITS architecture is typically constructed around four core modules[2, 10]: the Expert Model, which stores domain knowledge in structured formats such as semantic networks or rules; the Student Model, which tracks and evaluates students’ knowledge status; the Tutoring Model, which determines when and how to provide instructional interventions based on student and domain models; and the User Interface. Successful ITS projects like

Carnegie Tutor (Cognitive Tutor)[1] have demonstrated the effectiveness of this architecture in specific subjects such as mathematics and programming.

However, this classic four-module architecture with an integrated design also has limitations. The tight coupling between modules complicates system scalability and maintenance. The four-agent framework proposed in this study—comprising a planner, diagnostic module, instructor component, and Assessment Engine—represents a modernized and modular reinterpretation of traditional Intelligent Teaching Systems (ITS). By decoupling macro-level planning, micro-level diagnostics, and instructional execution responsibilities, we have enhanced the system’s flexibility and scalability.

2.2 Student modeling and knowledge representation

Student modeling is central to achieving personalized education. Traditional approaches primarily rely on quantitative analysis of students’ responses to extensive practice questions. For instance, Bayesian Knowledge Tracing (BKT)[4] uses hidden Markov models to infer students’ mastery probabilities of knowledge points. To capture more complex learning processes, subsequent research developed Deep Knowledge Tracing (DKT)[11], which employs recurrent neural networks (RNNs) to model students’ knowledge state evolution and handle more intricate skill dependencies. While these models excel in quantitative assessment, they face two major limitations: first, their heavy reliance on structured answer data makes them unsuitable for open-ended dialogue scenarios; second, they typically yield a single mastery score that fails to capture the multidimensionality of students’ cognitive states. Beyond knowledge tracing, the field of cognitive diagnosis models (CDMs) has focused on identifying fine-grained mastery of specific skills or attributes, but often relies on extensive, pre-designed assessments[14]. The C, A, S, R multidimensional cognitive model adopted in this study aims to address these shortcomings. By leveraging the analytical capabilities of large language models (LLMs) in qualitative dialogue analysis, it extracts richer and more interpretable cognitive states, thereby establishing a foundation for precise Socratic questioning.

2.3 Large language models and Socratic teaching

With the rise of large language models like ChatGPT, their direct use as AI tutors has become a research hotspot[7]. Leveraging their exceptional linguistic fluency and extensive global knowledge, LLMs outperform traditional Instructional Technology Systems (ITS) in simulating human dialogue. Many LLM-based systems also claim to enable "Socratic-style dialogues".

However, authentic Socratic teaching goes far beyond a series of simple "why" questions. It represents a highly dynamic and diagnostic-driven pedagogical artistry that requires mentors to not only accurately assess students’ current knowledge base and potential misunderstandings, but also to formulate well-timed questions that stimulate critical thinking without providing direct answers. Most existing LLM tutors struggle to consistently achieve this, as they typically lack an independent, structured student model. These systems tend to respond to the entire conversation history as context, resulting in implicit, unstable diagnoses that lack long-term

memory. Consequently, their instructional strategies often deviate from intended objectives, ultimately degenerating into basic question-and-answer bots. Some research has attempted to address this by pre-training models specifically on Socratic-style dialogues to improve their reasoning capabilities[16]. While such approaches can enhance the model’s stylistic performance, they still lack an independent, dynamic student model to guide the dialogue in real time. Our study, in contrast, argues that authentic Socratic teaching hinges not on a pre-trained style, but on a diagnosis-driven process. To address the hallucination issue in Large Language Models (LLMs) and ensure controllable knowledge sources, Retrieval-Augmented Generation (RAG) technology was developed[8]. In the education sector, RAG has been applied to build question-answering systems tailored for specific teaching materials. However, most implementations remain at the basic "question-answer" level, failing to deeply integrate RAG with structured teaching workflows.

2.4 Research positioning

In conclusion, although there has been a lot of research on the application of ITS and LLM in education, there is still a lack of a complete solution that can deeply integrate the flexibility of RAG, the reasoning ability of LLM and a structured framework designed specifically to support high-quality Socratic teaching.

This study aims to address this critical gap. We propose an innovative framework that decouples "student status analysis" from "dialogue generation" through a dedicated diagnostic agent. This agent constructs and maintains a robust, multi-dimensional student model, serving as a reliable instructional roadmap. Instructors then leverage this precise diagnostic report to generate targeted, thought-provoking Socratic-style questions. Our research pioneers an end-to-end intelligent teaching paradigm that transforms unstructured content input into structured, adaptive instructional outputs centered on Socratic dialogue.

3 System Architecture and Methodology

This chapter details the innovative intelligent tutoring system designed for dynamic and adaptive Socratic-style teaching. We begin by outlining the system’s multi-agent architecture, then delve into the internal algorithms and workflows of its core components: the Planner, Diagnoser, Tutor, and Assessment Engine.

3.1 Overall System Architecture

To overcome the limitations of monolithic ITS architectures[6], we propose a modular framework featuring four core intelligent agents working in synergy, a paradigm aligned with the trend of building agentic AI systems for education. The design adheres to the principle of separation of duties, with each agent assuming a distinct and pedagogically logical role: The Planner Agent preprocesses unstructured learning materials by dynamically constructing a structured knowledge base and generating a macro-level learning path aligned with the learner’s goals. The Diagnoser Agent performs real-time analysis of student–tutor interactions, maintaining and continuously updating a persistent, multidimensional model of the learner’s mastery across knowledge points. Building on this diagnostic feedback, the Tutor Agent orchestrates the instructional cycle, making pedagogical decisions and engaging the

student through high-quality Socratic dialogue. At critical stages, the Assessment Engine, invoked by the Tutor Agent, generates formative assessments that serve as “mastery validation checkpoints,” providing quantitative evidence of learning progress and reinforcing the closed-loop nature of the system.

3.2 PlannerAgent

The Planner Agent’s core mission is to break the “knowledge bottleneck” of traditional ITS by automatically converting unstructured documents into a structured teaching blueprint. This process is accomplished via two main stages inspired by the GraphRAG framework.[5].

3.2.1 Knowledge Extraction via a “Divide-and-Integrate” Paradigm. To process large documents while ensuring global consistency, we employ a “Divide-and-Integrate” approach. First, input documents are partitioned into semantically coherent text chunks, which are then vectorized and stored in a vector database for fast retrieval. The extraction process then unfolds in two phases:

- (1) **Map (Parallel Extraction):** Each text chunk is processed in parallel. A specialized LLM prompt, acting as a domain expert, extracts preliminary candidate knowledge points from the local context of each chunk.
- (2) **Integrate (Clustering & Refinement):** To address the redundancy and fragmentation from local extraction, we implement an intelligent refinement strategy. Candidate knowledge points are first vectorized and clustered using the Agglomerative Clustering algorithm. We chose this algorithm because it does not require pre-specifying the number of clusters, which is ideal for handling diverse, user-provided documents where the number of underlying topics is unknown. Subsequently, another LLM prompt, acting as a “knowledge base curator,” processes each cluster to merge, deduplicate, and enhance related entities, ultimately producing a clean and structured knowledge base in JSON format.

3.2.2 Learning Path Planning via a “Three-Stage Rocket” Algorithm. After the dynamic knowledge base is built, the system prompts the user to enter specific learning objectives. The planner then implements what we call a “three-stage rocket” multi-stage hybrid intelligent algorithm to ensure the efficiency and accuracy of path planning. The first stage of the algorithm is vector screening. It takes advantage of the efficiency of vector database to vectorize users’ learning objectives and quickly retrieve the Top N candidate knowledge points with the most semantic relevance from the complete knowledge base. This step greatly reduces the decision space. The second stage involves LLM selection. Since vector similarity may not fully capture logical connections in teaching content, we present the N candidate knowledge points identified through initial screening to an LLM acting as a “teaching expert”. This LLM is tasked with selecting essential subsets directly relevant to the user’s learning objectives and required for achieving those goals, thereby enabling deeper semantic and logical filtering. The third stage involves LLM ranking. Finally, the system presents this compact “essence” subset to the LLM using a thought-chain-guided prompt. This prompt explicitly instructs the LLM to first filter and

then sort the knowledge points, ultimately arranging them according to the teaching logic of progressing from basic to advanced concepts in order of difficulty. The process generates a logically coherent learning path composed of knowledge point IDs.

3.3 DiagnosticAgent

To provide precise guidance for high-quality Socratic dialogue, the Diagnoser Agent constructs and maintains an interpretable, multidimensional student model, moving beyond traditional single-score metrics like those from BKT or DKT. This aligns with recent efforts to create explainable student models using the reasoning capabilities of LLMs, sometimes referred to as ‘Chain-of-Thought Diagnosis’[9].

3.3.1 Multidimensional cognitive state model. We track student mastery across four dimensions, collectively forming their Cognitive State for each knowledge point:

- **C (Comprehension):** Mastery of conceptual knowledge (“what” and “why”).
- **A (Application):** Ability to apply knowledge to solve problems (“how”).
- **S (Strength):** The long-term stability of the memory trace for the knowledge point. This value decays over time based on the principles of the forgetting curve.
- **R (Recall):** The probability that the student can successfully retrieve the knowledge at a given moment. It is dynamically calculated based on memory strength (S) and the time elapsed since the last review (Δt). We model this using a standard exponential decay function:

$$R_t = e^{-\frac{\Delta t}{S}}$$

3.3.2 Diagnostic Process. The diagnostic process is initiated by the instructor in each teaching cycle. The system first retrieves the student’s historical mastery of the current knowledge point from persistent storage, defaulting to zero if no record exists. A context-rich “Cognitive Diagnosis Expert” prompt is then constructed, integrating the student’s historical mastery, ongoing dialogue history, and the knowledge points being taught, and submitted to a large language model (LLM) for analysis. The LLM synthesizes all information and returns structured JSON outputs that evaluate the student’s behavioral state (e.g., ENGAGED or STUCK) and update four cognitive dimensions. Returned data undergo rigorous format and range validation before being written back to persistent storage, completing a “read-analyze-update” diagnostic cycle.

Leveraging the reasoning capabilities of LLMs allows the system to extract richer, more interpretable cognitive states. While prior work on “Chain-of-Thought Diagnosis” enhances transparency of student reasoning, it often lacks persistent, multi-dimensional state representations. Our C, A, S, R model addresses this gap by maintaining and updating a structured, long-term cognitive profile for each knowledge point, providing a historically-informed foundation for precise Socratic questioning and supporting adaptive, memory-informed instruction.

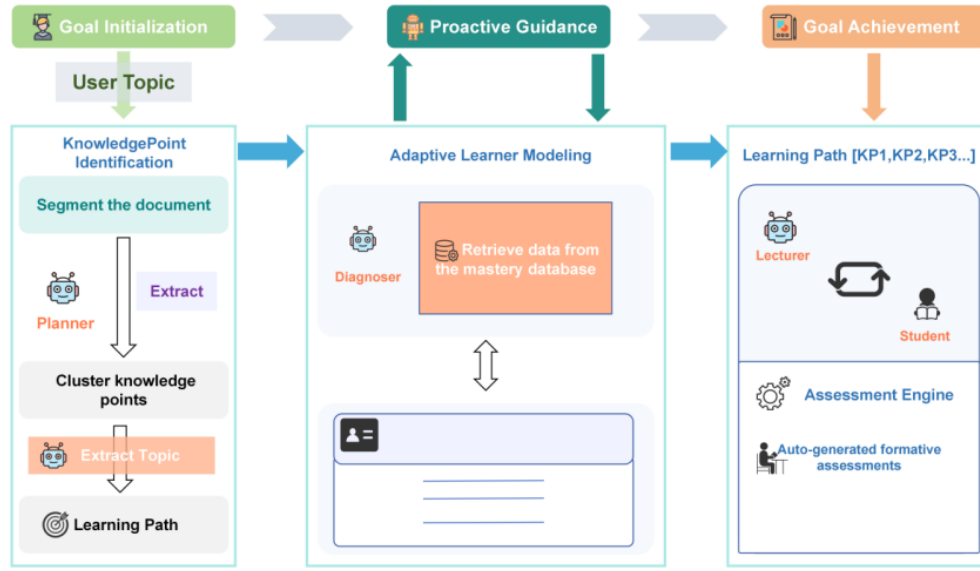


Figure 1: System overall architecture diagram

3.4 Tutor Agent

The Tutor Agent serves as the system's central instructional executor and "command center." Its primary role is to orchestrate the other agents and, guided by real-time diagnostics, engage the student in high-quality, adaptive Socratic dialogue. To achieve this, the Tutor Agent implements a robust, assessment-driven, two-mode teaching cycle.

3.4.1 Diagnosis-Driven Teaching and Assessment Cycle. The Tutor Agent's operation is governed by a dynamic cycle that alternates between two distinct modes, as illustrated in Figure 3.

- (1) **Teaching Mode:** This is the default operational state. In each turn of the dialogue, the Tutor Agent first requests an up-to-date diagnostic report from the Diagnoser Agent. This report provides a multi-dimensional assessment of the student's current understanding. Based on this diagnosis, the Tutor Agent synthesizes the student's state with predefined pedagogical strategies to construct a master prompt. This prompt then guides a large language model to generate the next Socratic-style response, ensuring the dialogue is tailored to the student's immediate learning needs.
- (2) **Assessment Mode (Mastery Verification Checkpoint):** To prevent the illusion of comprehension ("false mastery") and ensure that learning is robust, the system transitions into this mode when a key condition is met. The transition is triggered when the Diagnoser's report indicates that the student's Comprehension (C) and Application (A) scores for a given knowledge point have both surpassed a preset threshold (e.g., 0.8) for the first time. In this mode, the Tutor Agent invokes the **Assessment Engine** to dynamically generate a short, formative quiz focused

on the current knowledge point. The student must complete this assessment.

- If the student passes the quiz (e.g., achieves a score > 0.8), their mastery is considered verified, and the system advances to the next topic in the learning path.
- If the student fails, the system reverts to **Teaching Mode**, providing targeted supplementary instruction based on the weaknesses identified in the quiz before allowing another mastery attempt.

This closed-loop design, which tightly integrates formative, dialogue-based diagnosis with summative, quiz-based verification, ensures a rigorous and effective teaching process.

4 Experimental Design and Evaluation

To comprehensively validate the effectiveness of our proposed Multi-Agent Adaptive Teaching Framework (OurSystem), this chapter designs and implements a systematic experimental protocol. We first define core evaluation tasks, then sequentially outline experimental configurations and assessment methodologies specifically targeting teaching adaptability and instructional planning capabilities.

4.1 Core Evaluation Tasks

The core innovation of OurSystem lies in its dynamism and adaptability, so the evaluation focuses on the following two capabilities:

- **Adaptive Tutoring Capability:** The ability of the evaluation system to dynamically adjust the teaching dialogue during the teaching process based on real-time diagnosis, which is mainly completed by the collaboration of the diagnosticians and the Tutors.
- **Teaching planning ability (Planning Capability):** The ability of the evaluation system to build a high-quality knowledge

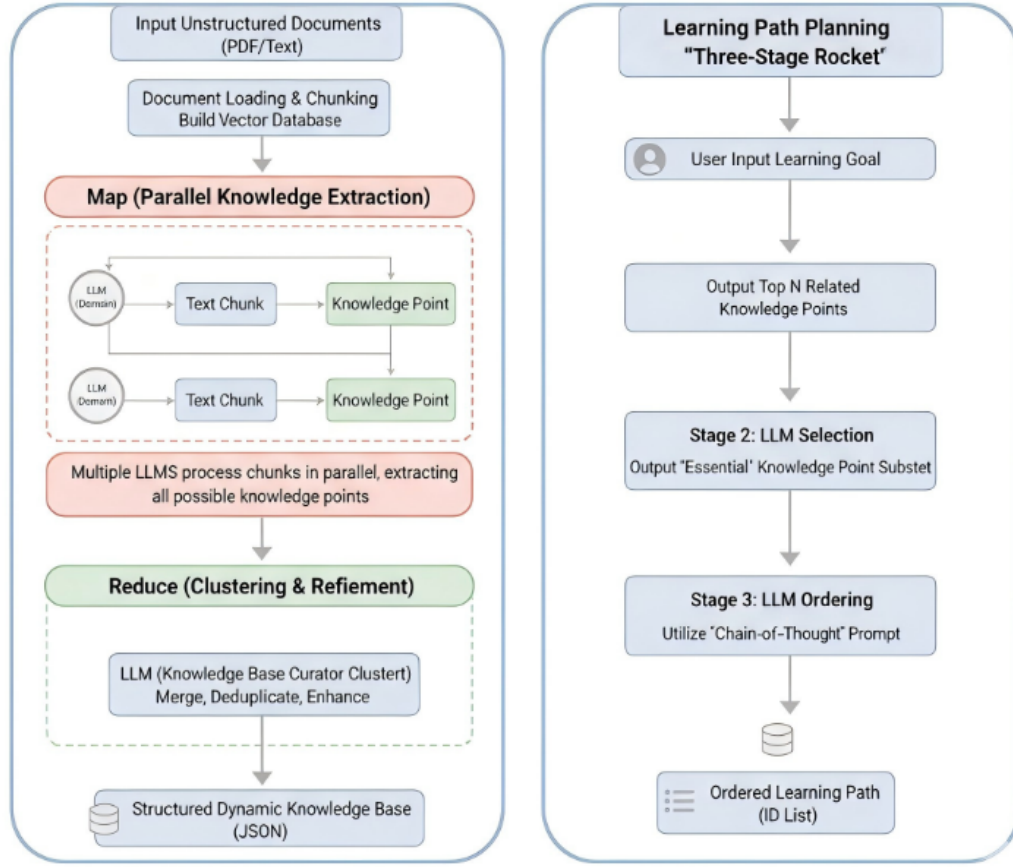


Figure 2: The Architecture and Workflow of the PlannerAgent

base and reasonable learning path from unstructured documents before teaching begins, which is mainly responsible by the planner.

4.2 Experimental setup of teaching adaptability

To rigorously evaluate the adaptive tutoring capabilities of our proposed system (OurSystem), we conducted a comparative experiment against a strong baseline. The test set consisted of 100 independently sampled, single-turn dialogue scenarios from the SocraTeach dataset, each including the complete dialogue history and the student’s most recent response to provide realistic tutoring context. OurSystem, configured with the Gemma3-27B model, generated responses through a two-step procedure: the Diagnoser Agent first produced a student mastery report from the dialogue history, which the Tutor Agent then used, along with predefined instructional decision rules, to generate an adaptive response. The baseline, employing GPT-3.5-Turbo, followed a conventional non-diagnostic approach, generating responses directly from a generic Socratic-style prompt without leveraging a structured student model. Evaluation employed a dual approach combining automated and human assessments, both conducted under blind conditions. An AI

jury comprising state-of-the-art large language models (ChatGPT-4o, Gemini-2.5, and Claude-3.5) performed large-scale evaluation across all scenarios, while human evaluation was conducted on a randomly selected subset of 20 scenarios by a team of 10 trained university students following the same procedure as the AI jury. In both assessments, responses were compared across four pedagogical dimensions—instructional adaptability, Socratic style, encouragement, and clarity—with evaluators providing comparative judgments and brief justifications.

4.3 Planning Capability Evaluation

To quantitatively evaluate the effectiveness of this study in instructional planning, we designed a teaching blueprint construction experiment. This experiment compares the performance of our proposed PlannerAgent (OurSystem) against a Baseline method to assess the core capability of developing teaching blueprints from unstructured texts.

The Baseline method represents a common, non-decoupled approach, designed to accomplish the planning task in a single, end-to-end step. It operates by feeding the entire learning document

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1 Prompt: You are a top-level cognitive diagnosis expert and AI tutor.
  Please analyze the dialogue between the student and tutor and assess
  the student's mastery of a specific knowledge point.
2 Diagnostic Context:
3 ...
4 Student Historical Mastery (0.0-1.0):
5 - C (Comprehension): {}
6 - A (Application): {}
7 - S (Strength): {}
8 - R (Recall): {}
9 ...
10 "behavioral_state": "",
11 "cognitive_state": {
12   "C": 0.0-1.0,
13   "A": 0.0-1.0,
14   "S": 0.0-1.0,
15   "R": 0.0-1.0
16 },
17 "analysis_notes": "Brief explanation"
18 }
19 ...

```

Figure 3: Cognitive Diagnosis LLM Prompt

and the user-specified learning objective into a single, comprehensive prompt for a general-purpose large language model (GPT-3.5-Turbo). The model is then instructed to perform both knowledge extraction and path planning simultaneously. In contrast, OurSystem utilizes the specialized "divide-and-conquer-integrate" and "Three-Stage Rocket" architecture as detailed in Section III.

To achieve a more precise analysis, we decomposed the overall planning ability into two independent core subtasks for evaluation: knowledge base construction and learning path planning.

4.4 Data sets and tasks

This study utilizes a high-quality test set —the High School Mathematics Core Concepts Test Set, which we manually constructed and annotated. The dataset encompasses five core mathematical domains (including quadratic functions, spatial vectors, and fundamentals of probability theory), with 2-3 specific learning objectives defined for each domain. It contains 13 distinct test samples designed to assess understanding across these subjects. For each document and learning objective, we manually annotated: 1. Ideal list of core knowledge points (for evaluating the construction of knowledge base) 2. Ideal learning path (for evaluating path planning), and finally formed 13 independent test samples.

4.5 Assessment indicators

In order to ensure the robustness and fairness of the evaluation, we adopt a set of semantic similarity based evaluation indicators to deal with the diversity of output from large language models.

- Evaluation metrics for knowledge base construction: Precision, Recall, and F1-Score: These metrics measure task performance by assessing semantic overlap between extracted

knowledge points and ideal knowledge point lists. We employ the bge-m3 vector model to calculate cosine similarity between knowledge point descriptions and utilize the Hungarian algorithm to identify global optimal matches, preventing misjudgments caused by model generalization or paraphrasing. A successful match is confirmed when the semantic similarity between two knowledge points exceeds the threshold of 0.80.

- Evaluation Metrics for Learning Pathway Ranking: Path Jaccard Similarity (PJS): This metric measures the model's ability to filter knowledge points during pathway selection. It calculates the Jaccard similarity coefficient between the knowledge point set contained in the model's sorted pathway and the ideal pathway's knowledge point set. Kendall's Tau (τ): This metric evaluates task two's performance by comparing the consistency of knowledge point order between the model's ranking results and the ideal pathway, assessing the logical coherence of the teaching pathway arrangement.

4.6 Experimental Procedure

We conducted automated evaluations on 13 test samples through the following process:

Knowledge Base Construction Assessment: For each sample, we first compiled all descriptive texts from its ideal knowledge point list (ground_truth_kps) into a simulated learning document. This document was then fed into the PlannerAgent, which executed its build_knowledge_base method to generate a predicted knowledge base. Semantic matching between the predicted knowledge points and the sample's actual knowledge points (ground_truth_kps) was performed, with precision, recall, and F1-Score metrics calculated accordingly.

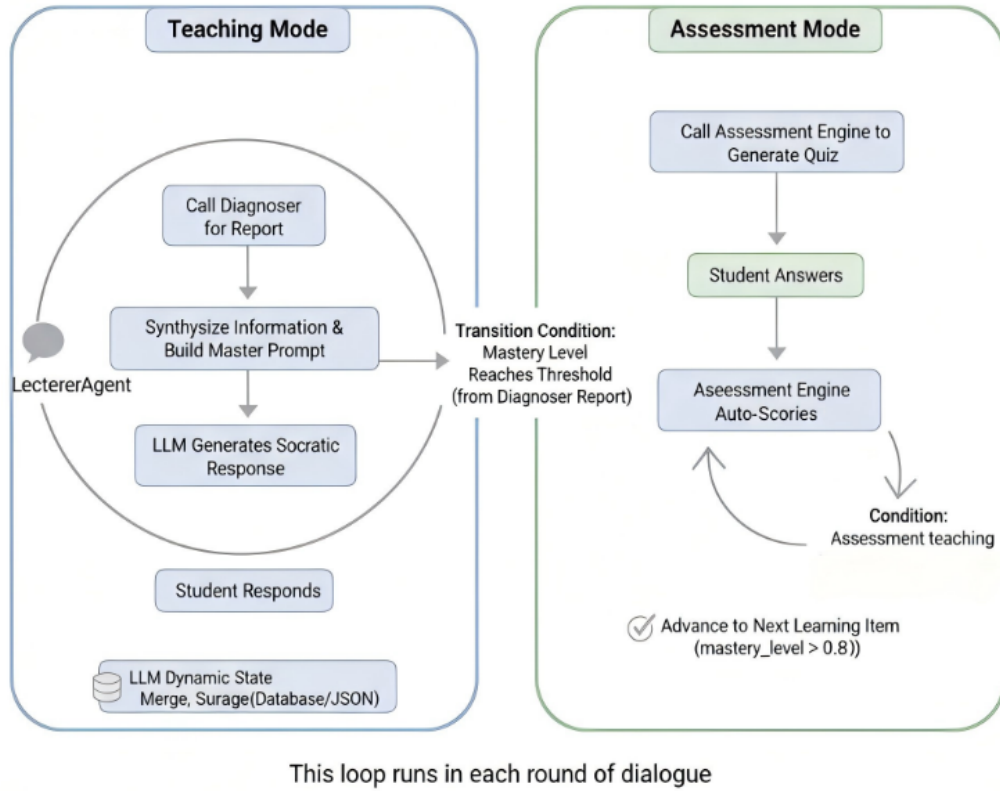


Figure 4: The Architecture and Workflow of the LecturerAgent

Learning Pathway Ranking Evaluation: For each sample, we obtain its ideal learning path (*ground_truth_path*) as unordered knowledge input along with corresponding learning objectives. This "perfect" knowledge set is fed into the *PlannerAgent*'s ranking module to generate an ordered learning pathway. The model's output ranking path is then compared with the ground truth path to calculate path content overlap rate and Kendall's rank correlation coefficient.

By decoupling experimental tasks, this study enables independent and precise evaluation of *PlannerAgent*'s performance in two core instructional planning aspects: knowledge extraction and logical sequencing.

5 Experimental Results and Analysis

This section presents the empirical evaluation of the proposed framework, *OurSystem*. We first analyze the results of the adaptive tutoring evaluation, followed by an assessment of the instructional planning capabilities.

5.1 Results of Adaptive Tutoring Evaluation

The adaptive tutoring capability of *OurSystem* was evaluated against the Baseline through both automated AI Judge assessments

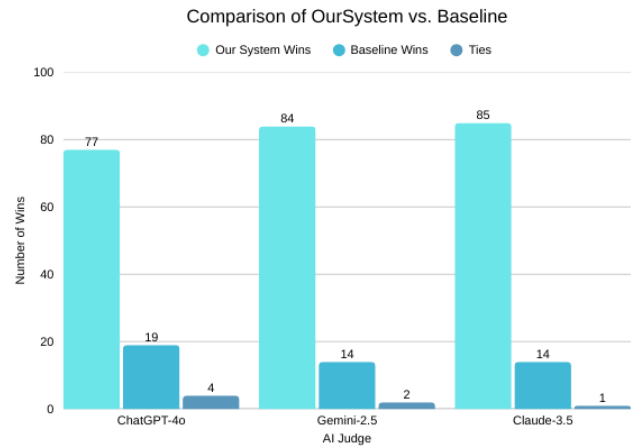


Figure 5: AI Judge Evaluation Results: Comparison of Win Rates

and manual human evaluations. The results from both methods affirm the superiority of our diagnosis-driven approach.

Table 1: Dimensional Analysis of OurSystem vs. Baseline (Mean Values)

Dimension	OurSystem (Gemma3-27B)	Baseline (GPT-3.5-Turbo)
Socratic Style	4.30	3.45
Teaching Effectiveness	4.27	3.28
Encouragement	4.30	4.37
Clarity	4.50	4.42
Overall	4.34	3.88

5.1.1 AI Judge. In a blind A/B test conducted by an AI review panel, OurSystem demonstrated a significant advantage over the Baseline. As shown in Figure 4, OurSystem achieved high win rates across all three judges: 77% (vs. 19% for Baseline) as judged by ChatGPT-4o, 84% (vs. 14%) by Gemini-2.5, and 85% (vs. 14%) by Claude-3.5.

- In the chatgpt-4o model, OurSystem was rated as superior 77% of the time, while Baseline won 19% of the time.
- In the gemini-2.5 model, OurSystem was rated as superior 84% of the time, while Baseline won 14% of the time.
- In the claude-3.5 model, OurSystem was rated as superior 85% of the time, while Baseline won 14% of the time.

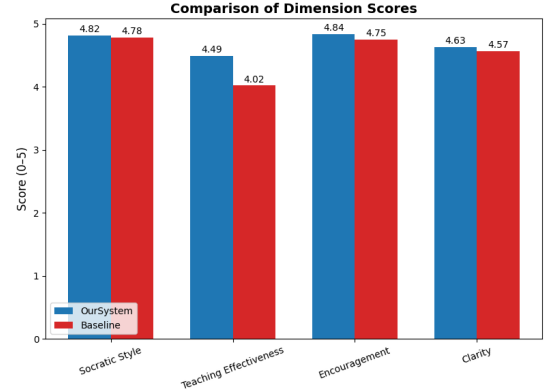
5.1.2 Qualitative Case Study. To complement the quantitative data, the following case study illustrates the practical difference in instructional quality.

- **Scenario:** A student is solving a multi-step multiplication word problem. After some initial confusion, they make a final calculation error in addition.
- **Student’s Final Statement:** "...so it’s 6 for Adam, plus 18 for James, plus 72 for Rubert, which makes 93.9 candies."
- **Baseline’s Response:** "Great job! You did an excellent job working through the multiplication and addition... Now, let’s try one more question to challenge ourselves..."
- **OurSystem’s Response:** "That’s very close! You’ve done all the right calculations. However, we can’t have parts of candies, can we? Let’s double-check your addition: $6+18+72$. Can you carefully re-add those numbers, making sure you haven’t made a small mistake? Sometimes a fresh look is all we need!"

The analysis is stark: the Baseline, lacking diagnostic insight, offers incorrect praise and moves on, reinforcing the student’s error. In contrast, OurSystem identifies the specific error (addition, not multiplication) and provides a targeted, Socratic prompt to guide the student toward self-correction. This demonstrates a fundamentally more effective and adaptive teaching approach.

5.1.3 Human Evaluation. The results from the human evaluation align with the AI Judge findings, reaffirming the superiority of OurSystem. Out of 191 valid pairwise comparisons, OurSystem secured 112 wins compared to the Baseline’s 52 (27 ties). This constitutes a win rate of 68.3%, which is statistically significant and indicates a clear preference among human evaluators.

The dimensional analysis (Figure 5) reveals a crucial insight: OurSystem demonstrated exceptional performance in Teaching Effectiveness. Its mean score (4.49) was not only significantly higher than the Baseline’s (4.02), Its mean score (4.49) was not only significantly higher than the Baseline’s (4.02), but the variance of its

**Figure 6: Human Evaluation Results: Comparison of Dimension Scores**

Teaching Effectiveness scores was also substantially lower (0.60 vs. 1.73). This suggests that OurSystem is not only more effective but also more consistent and reliable in its teaching quality.

Interestingly, a nuanced difference emerged between AI and human evaluations. While the AI judges rated OurSystem higher on Socratic Style, human evaluators gave the OurSystem a narrow lead (4.82 vs. 4.78). We hypothesize that human evaluators may associate the linguistic fluency of GPT-3.5-Turbo with a positive "style," whereas the AI judges placed more weight on the logical coherence and adaptability driven by our diagnostic module. Nevertheless, OurSystem’s decisive victory on the most critical metric—Teaching Effectiveness—validates the core value of its diagnosis-driven architecture.

5.1.4 Analysis of Planning Performance. This quantitative evaluation provides strong empirical support for our core hypothesis: decoupling "knowledge discovery" from "logical sequencing" is a highly effective strategy for instructional planning. The baseline’s near-total failure in knowledge base construction reveals that knowledge discovery from unstructured documents is the primary bottleneck in the end-to-end planning process, a challenge that standard retrieve-and-rank architectures are ill-equipped to handle.

Our PlannerAgent successfully addresses this bottleneck. Its "divide-and-integrate" paradigm for knowledge extraction demonstrates a profound advantage in building a comprehensive and accurate knowledge foundation. Subsequently, the "Three-Stage Rocket" algorithm for path planning refines this foundation into a pedagogically sound sequence that surpasses even a strong ranking baseline. The near-human-level performance of OurSystem validates that the synergy between a robust algorithmic structure and the reasoning capabilities of large models like Gemma3-27B is crucial for solving complex, real-world instructional design problems.

6 Discussion

6.1 Principal Findings

This study set out to answer two primary research questions, and the experimental results provide clear answers.

Table 2: Performance Evaluation Metrics of Planning Capability

Metric	OurSystem(Gemma3-27B)	Baseline(GPT-3.5-Turbo)
Knowledge Base Construction		
Precision	0.4544	0.0308
Recall	0.6059	0.0308
F1-Score	0.5154	0.0308
Learning Path Planning		
Path Jaccard Similarity	0.9904	0.6691
Kendall's Tau	0.9300	0.7396

For RQ1 (Instructional Planning), the experiments demonstrated that the proposed PlannerAgent, with its "divide-and-integrate" and "three-stage rocket" algorithms, significantly outperforms a monolithic baseline. It achieves superior recall in knowledge base construction and near-perfect logical consistency in learning path planning. This validates that our dynamic planning framework can effectively construct high-quality teaching blueprints from unstructured documents.

For RQ2 (Instructional Adaptability), the A/B blind testing revealed that the complete system, integrated with the DiagnoserAgent, achieved an average win rate of 82% against a non-diagnostic baseline in AI evaluations. This confirms that the diagnosis-driven architecture is highly effective at enhancing the quality of Socratic dialogue. The key mechanism is the real-time analysis of student cognition, which enables the Tutor Agent to generate personalized and adaptive feedback, thereby realizing true differentiated instruction.

6.2 Implications for Intelligent Tutoring Systems

The findings of this study offer several important implications. First, we provide strong evidence that a dynamic diagnostic mechanism is a core prerequisite for truly adaptive teaching, enabling AI tutors to transition from generic "chatbots" to personalized "coaches" capable of targeted intervention. Second, the success of the PlannerAgent validates the hybrid intelligence paradigm—combining the efficiency of vector search with the deep reasoning of LLMs—as a viable and effective approach to solving complex instructional planning problems. Finally, our work highlights both the potential and the challenges of AI-driven automated assessment; while it enables large-scale evaluation, it also underscores the urgent need for fair and stable metrics to measure the quality of complex teaching interactions.

6.3 Limitations and Future Work

Despite the promising results, this study has several limitations that directly inform our agenda for future research.

First, our evaluation's primary limitation is the absence of a long-term study on learning effectiveness with real human learners. The current assessment relies on automated metrics and short-term interactions, which serve as proxies but do not directly measure

crucial outcomes like knowledge retention or critical thinking development. Future work will prioritize large-scale, controlled experiments in authentic educational settings to measure and validate long-term learning gains.

Second, the semantic similarity threshold of 0.80 was selected based on heuristics and lacks rigorous validation. Since evaluation scores are sensitive to this parameter, future work should include a sensitivity analysis and calibrate the threshold against a manually annotated validation set to establish a more empirically justified benchmark.

Finally, the framework's generalizability is currently limited to mathematics. Extending it to other disciplines like physics or computer science will be a key focus of future work, requiring adaptation of the cognitive and pedagogical models. Additionally, the system's performance is inherently dependent on the capabilities of the underlying large language models.

Building on this foundation, we will also explore more sophisticated student models incorporating metacognitive and emotional factors, while carefully addressing the associated ethical considerations of data privacy and transparency.

7 Conclusion

This paper introduced a diagnosis-driven, multi-agent framework for adaptive Socratic tutoring. By decoupling planning, diagnosis, and instruction into specialized agents, our system can dynamically create teaching blueprints from unstructured content and deliver personalized, effective Socratic dialogues. Our experiments confirm that this diagnosis-driven architecture significantly outperforms non-adaptive baselines in both instructional planning and teaching quality. This work provides a robust architectural blueprint for the next generation of intelligent tutoring systems that are truly adaptive and responsive to the needs of individual learners.

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